

PROJECT DEVELOPMENT PHASE



Functional Features Overview

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 NAME
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 PROJECT
OptiCrop – Smart
Agricultural
Production
Optimization Engine

 MAX MARKS
5 Marks

Overview

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

★ Feature Checklist

FEATURE	DESCRIPTION	MODULE/COMPONENT	STATUS	CONTRIBUTION
Soil & Climate Input Form	Allows users to enter N, P, K, temperature, humidity, pH, and rainfall readings.	Frontend (HTML/CSS)	Done	High
Input Validation	Validates user input before processing.	Flask Backend	Done	High
Data Preprocessing	Cleans and transforms input data into the required format.	Preprocessing Module	Done	High
Crop Recommendation	Predicts the most suitable crop using the trained ML model.	Machine Learning Model	Done	High
Result Display	Displays the recommended crop and a short description to the user.	Flask Application	Done	High
Trained Model Loading	Loads the saved machine learning model (model.pkl) for prediction.	Model Loader	Done	High
Responsive User Interface	Provides a user-friendly interface accessible through a web browser.	HTML/CSS/JavaScript	Done	High
Error Handling	Handles invalid or incomplete user input gracefully.	Flask Backend	Done	High